

Sources & Text Books

- David Tong's lecture notes QFT
- Sidney Coleman's lecture notes & videos
- Mandl & Shaw QFT
- Weinberg (not for beginners) → most rigorous
- Peskin & Schroeder Ch 2-4
- Lee (intuitive)

Concise previous requirements

Lagrangian Mechanics $\mathcal{L}(\dot{q}_a, q_a, \tau)$

$$S = \int_{\tau_1}^{\tau_2} \mathcal{L} d\tau$$

The true path is $\delta S = 0$

$$\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) = \frac{\partial \mathcal{L}}{\partial q_i}$$

↙
Differential
Equation

Noether's Theorem

Considere uma transformação infinitesimal

$$\delta q_a = \chi(\dot{q}, q, \tau)$$

isso é uma simetria para $\delta \mathcal{L} = 0$ para todos os caminhos

Prova: considere uma deformação arbitrária

$$\delta q_a$$

$$\delta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial q} \delta q + \frac{\partial \mathcal{L}}{\partial \dot{q}} \delta \dot{q} \quad \frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) = \frac{\partial \mathcal{L}}{\partial q}$$

$$= \underbrace{\left(\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) \right)}_{0 \text{ para True Path}} \delta q + \underbrace{\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right)}_{\mathcal{Q}} \delta q$$

$$\delta \mathcal{L} = \frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) \delta q \iff \delta \mathcal{L} = \dot{\mathcal{Q}}$$

Se que definirmos $\delta q = \chi(q, \dot{q}, \tau)$ com $\delta \mathcal{L} = 0$

$\frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{q}} \right) \chi(q, \dot{q}, \tau)$ então temo que $\dot{Q} = 0$
 assim Q é uma quantidade conservada

Formalismo Hamiltoniano

$$P^\alpha = \frac{\partial H}{\partial \dot{q}_\alpha} \quad H = T + K$$

$$H(P, q) = P^\alpha \dot{q}_\alpha - L \quad \text{com } L = \left(\frac{mv^2}{2} - U \right)$$

$$\dot{q}_\alpha = \frac{\partial H}{\partial P^\alpha}, \quad \dot{P}^\alpha = - \frac{\partial H}{\partial q_\alpha}$$

Poisson Bracket

Sejam duas funções $f(q, P)$ e $g(q, P)$

$$\{f, g\} = \frac{\partial f}{\partial q_\alpha} \frac{\partial g}{\partial P^\alpha} - \frac{\partial f}{\partial P^\alpha} \frac{\partial g}{\partial q_\alpha}$$